Sukhyun Han

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2066, Seobu-ro, Jangan-gu, Suwon-si, Republic of Korea

Research Interests

High-Performance AI Accelerator & NPU Architecture
- Workload-aware dataflow and microarchitecture optimization for emerging AI workloads
- HW/SW co-design to improve compute utilization and mitigate memory bottlenecks

Next-Generation Memory-Based AI Computing Systems
- Scalable system architectures integrating NPUs with next-generation memory such as HBM and 3D DRAM
- Full-stack system modeling and cross-layer performance optimization for LLMs

Education

Ph.D. Program
Sungkyunkwan University (SKKU), Suwon, South Korea — Sep. 2025 ~ Present
Department of Electrical and Computer Engineering (Advisor: Seokin Hong)

Master's Degree
Sungkyunkwan University (SKKU), Suwon, South Korea — Mar. 2024 ~ Aug. 2025
Department of Semiconductor Convergence Engineering (Advisor: Seokin Hong)

Bachelor's Degree
Sungkyunkwan University (SKKU), Suwon, South Korea — Mar. 2019 ~ Feb. 2024
Department of Electronic and Electrical Engineering

Skills and Techniques

Programming Languages
- Digital system Design: Verilog, System Verilog
- Software Development: C, C++, Python, MATLAB, CUDA on Linux Environment

Tools
- Synopsys Design Compiler, Cacti-7.0, Verdi
- Vivado FPGA Suite, Vitis
- Cadence Virtuoso Suite, PSpice, LTspice

Honors and Awards

Best Paper Award, ITC-CSCC 2025 (co-author paper "Outlier Matters") — Jul. 2026

Graduate Academic Excellence Scholarship, Sungkyunkwan University — Sep. 2025 ~ Aug. 2027

Outstanding Research Award from SKKU Graduate School (Master's Course) — Aug. 2025

AI Semiconductor design Competition (Popularity Award) — Feb. 2023

Deep learning Hardware Design Competition (4th place prize) — Jun. 2022

First Author Publications

Zebra: Leveraging Diagonal Attention Pattern for Vision Transformer Accelerator

- 2025 Design, Automation & Test in Europe Conference & Exhibition (DATE 2025)
- **Sukhyun Han**, Seongwook Kim, Gwangeun Byeon, Jihun Yoon, Seokin Hong
- This paper proposes a novel accelerator for ViT. This paper reveals that the self-attention map of ViT models consistently exhibits diagonal attention patterns, enabling pattern-aware pruning. To leverage these sparse diagonal patterns, this paper proposes novel hardware architecture that only compute necessary vector operations in attention mechanism.

Empirical Performance Analysis of Hybrid Bonding based 3D DRAM under LLM Inference

- The 41th International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC 2026)
- **Sukhyun Han**, Seokin Hong
- This paper quantifies the performance gap of hybrid-bonded 3D DRAM under LLM inference. It identifies three Bank-per-I/O bottlenecks: capacity overhead from data replication, NoC congestion from NUMA, and row-buffer conflicts from random KV-cache accesses.

Co-Author Publications

Avalanche: Optimizing Cache Utilization via Matrix Reordering for Sparse Matrix Multiplication Accelerator

- 2025 International Symposium on Computer Architecture (ISCA 2025)
- Gwangeun Byeon, Seongwook Kim, Hyungjin Kim, **Sukhyun Han**, Jinkwon Kim, Prashant Nair, Taewook Kang, Seokin Hong

Leveraging Mobile Processors for ISAR Image Generation and Classification in Radar Platforms

- IEEE Access (Volume 13, 2025)
- Seongwook Kim, **Sukhyun Han**, Woojin Cho, Youngjae Choi, Seungeui Lee, Youngseok Bae, Seokin Hong

Lupin: Spatial Resource Stealing with Outlier-First Encoding for Mixed-Precision LLM Acceleration

- 2026 Design, Automation & Test in Europe Conference & Exhibition (DATE 2026)
- Taein Kim, **Sukhyun Han**, Seongwook Kim, Gwangeun Byeon, Jungmin Lee, Seokin Hong

Outlier Matters: A Statistical Analysis of LLM Tensor Distributions and Quantization Effects

- The 40th International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC 2025)
- Taein Kim, Seongwook Kim, **Sukhyun Han**, Woojin Cho, Youngjae Choi, Youngseok Bae, Seokin Hong

Evaluating Performance of Modern GPU with Partitioned Last-Level Caches

- The 40th International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC 2025)
- Jihun Yoon, Joonseong Hwang, **Sukhyun Han**, Sungbin Jang, Yoonho Jang, Seokin Hong

Projects

Research Fellow Aug. 2023 - Present
Computer Architecture and Systems Lab – COMPASS LAB SKKU (Prof. Seokin Hong)
- Teaching Assistant for Semiconductor System Engineering courses.
- Maintain and Manage GPU A100 Computing server.

Scalable NPU Architecture Design based on Next Generation Memory Oct. 2025 - Present
COMPASS Lab
- In collaboration with Samsung Electronics Inc.
- Research and develop a scalable NPU architecture based on next-generation memory technologies for large-scale AI workloads. Integrated multi-core NPUs with 3D IC DRAM to minimize data access latency, significantly maximizing the efficiency of compute and memory integration.

PMBIST Design for Next Generation HBM Memory Jul. 2025 - Present
COMPASS Lab
- In collaboration with Samsung Electronics Inc.
- Develop a Programmable Memory Built-In Self-Test (PMBIST) architecture to address the complex testing requirements of next-generation HBM. My work included design of RISC-V based tiny core and user SDK to significantly enhance test pattern flexibility and execution efficiency.

AI Semiconductor Innovation Research Center Jul. 2025 - Present
COMPASS Lab
- Visiting Research at Samsung Electronics Inc.
- Conduct on-device AI performance optimization and advanced NPU architecture research by analyzing Samsung's proprietary Exynos NPU SDK (EHT/ELT) workflows.

Comprehensive HW/SW Solution for Edge Self Supervised Learning Sep. 2023 - Present
COMPASS Lab
- Design NPU Microarchitecture including ISA and the programming model integrated RISC-V Instruction Set Simulator.
- My work includes managing docs of Spike Simulator extension SDK and contributing to the development of RISC-V based extended Vector ISA for edge NPU specialized for Self-Supervised Learning model.

NPU Technology for Accelerating Ladar-based Object Detection Sep. 2023 - Feb. 2025
COMPASS Lab
- In collaboration with Hanwha Systems Inc.
- Evaluated the performance of Mobile AP embedded with NPU for ISAR-based object detection and classification task. My work included implementation of efficient ISAR-based algorithms on Mobile AP and design the control application.

Samsung Computer Engineering Challenge 2023 Aug. 2023 - Nov. 2023
Personal Project
- Optimized LLaMa2 model Inference without accuracy drop.
- My work included implementing multi-GPU parallelization and advanced dataset sorting with dynamic padding reduction to minimize redundant computations.

Robust BGR and LDO Design against PVT Variations Jun. 2022 - Aug. 2022
CIRCUITS Lab in SKKU (Undergraduate Research Program)
- Designed and evaluated AMP, BGR, and LDO circuits meeting specifications using 180nm TSMC process.
- My work included Improving and analyzing performance for PVT variation robustness by utilizing Thick PMOS and Cascode PMOS devices.

Implementation of Neural Network Computation using Low-Power CIM Circuit Mar. 2023 - Jul. 2023
Undergraduate Thesis
- Undergraduate Poster session at the 38th International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC 2023).
- My work included Implementation an energy-efficient Computing-in-Memory circuit based on 8T SRAM by incorporating a Layer Recycling technique.

Real Time Face Recognition using MTCNN and FaceNet with TensorRT Sep. 2021 - Dec. 2021
Undergraduate Course Project

- Optimized MTCNN and FaceNet algorithms on Jetson Nano using CUDA for GPU/CPU co-processing.
- My work included real-time multi-face recognition with K-NN algorithm and enabled new face registration functionality, enhancing accuracy.

Patents

Applications

- [KR] METHOD AND APPARATUS FOR VISION TRANSFORMER ACCELERATION USING STRIPED DIAGONAL PRUNING, Seokin Hong, **Sukhyun Han**, Seongwook Kim, Gwangeun Byeon, Jihun Yoon, 10-2026-0002953, 2026.01
- [KR] COMPUTING DEVICE OF SUPPORTING ACTIVATION OUTLIER HANDLING AND BUDGET-AWARE ENCODING, AND OPERATING METHOD THEREOF, Kukbyung Kim, Seokin Hong, Taein Kim, Younho Jeon, **Sukhyun Han**, Yoonho Jang, Kiwoong Kim, 10-2026-0151041, 2026.08

Experiences

SKKU Teaching Assistant

- SSE2029, Data Structures and Algorithms Fall, 2024
- SSE2036, Computer Architecture Design Spring, 2025
- SSE2029, Data Structures and Algorithms Fall, 2025
- EEE2017, Electrical and Electronic Programming Practice Spring, 2026